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EXERCICE B1 – 2025

$$f(x; \lambda) = \lambda e^{-\lambda x}, \quad x \in [0, \infty].$$

(a) Characteristic function

$$\begin{aligned}\varphi(t) &= \mathbb{E} [e^{1tx}] = \int_0^\infty e^{1tx} f(x; \lambda) dx = \int_0^\infty e^{1tx} \lambda e^{-\lambda x} dx = \lambda \int_0^\infty e^{(1t-\lambda)x} dx \\ &\Rightarrow \int_0^\infty e^{-(\lambda-1t)x} dx = \frac{\lambda}{\lambda - t} \\ &\Rightarrow \varphi(t) = \frac{\lambda}{\lambda - 1t}.\end{aligned}$$

(b) For two random variables X and Y both distributed as:

$$\begin{aligned}x &\sim f_x(x; \lambda) = \lambda e^{-\lambda x}, x \in [0, \infty] \text{ and} \\ y &\sim g(y; \lambda) = \lambda e^{-\lambda y}, y \in [0, \infty], \text{ the p.d.f. of } z = x + y \text{ is given}\end{aligned}$$

by the convolution of both p.d.f.s:

$$\begin{aligned}f(z; \lambda) &= \int_{-\infty}^\infty f_x(x; \lambda) g(z - x; \lambda) dx = \int_0^z f_x(x; \lambda) g(z - x; \lambda) dx \\ &= \int_0^z \lambda e^{-\lambda x} \cdot \lambda e^{-\lambda(z-x)} dx = \lambda^2 \int_0^z e^{-\lambda(x+z-x)} dx \\ &= \lambda^2 \int_0^z e^{-\lambda z} dx = \lambda^2 e^{-\lambda z} \int_0^z dx = \lambda^2 e^{-\lambda z} [x]_0^z = \lambda^2 z e^{-\lambda z}. \\ &\Rightarrow f(z; \lambda) = \lambda^2 z e^{-\lambda z}, \text{ with } z \geq 0.\end{aligned}$$

(c) Let $X_1, X_2, \dots, X_N$ be M exponentially distributed r.v.s and $S = \sum_{i=1}^N X_i$, i.e. $\forall i \in \{1, \dots, N\}$, $f_{X_i}(x) = \lambda e^{-\lambda x}$, $x \geq 0$. Base case: $M = 1 \Rightarrow f_{X_1}(x) = \lambda e^{-\lambda x}$ and $f(s; \lambda) = \frac{\lambda^1 s^{1-1} e^{-\lambda s}}{(1-1)!} = \lambda e^{-\lambda s}$, $s \geq 0$ matches $f_{X_1}(x) = f(s; \lambda)$

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Induction hypothesis: Assume that for some $N > 1$

$$f_M(s) = \frac{\lambda^N s^{N-1} e^{-\lambda s}}{(N-1)!}, s \geq 0$$

is the distribution of the sum of N such exponential random variables. Induction step.
let $S_{N+1} = S_N + X_{N+1}$. Using convolution, we have:

$$\begin{aligned}
 f_{N+1}(s) &= \int_0^s f_N(s)f_{X_{N+1}}(s-\mu)d\mu = \int_0^s \frac{\lambda^N \mu^{N-1} e^{-\lambda s}}{(N-1)!} \lambda e^{-\lambda(s-\mu)} d\mu \\
 &= \frac{\lambda^{N+1} e^{-\lambda s}}{(N-1)!} \int_0^s \mu^{N-1} e^{-\lambda(\mu+s-\mu)} d\mu = \frac{\lambda^{N+1} e^{-\lambda s}}{(N-1)!} \int_0^s e^{-\lambda \mu} d\mu \mu^{N-1} \\
 &= \frac{\lambda^{N+1} e^{-\lambda s}}{(N-1)!} \int_0^s \mu^{N-1} d\mu = \frac{\lambda^{N+1} e^{-\lambda s}}{(N-1)!} \left[\frac{\mu^N}{N} \right]_0^s \\
 &= \frac{\lambda^{N+1} s^N e^{-\lambda s}}{N!} \\
 \Rightarrow f_{N+1}(s) &= \frac{\lambda^{N+1} s^N e^{-\lambda s}}{N!}
 \end{aligned}$$

Thus, $\forall N \in \mathbb{N}, f_N(s; \lambda) = \frac{\lambda^N s^{N-1} e^{-\lambda s}}{(N-1)!} \quad s \geq 0$. (d) Let's consider some dataset $\{x_1, x_2, \dots, x_N\}$. The likelihood is given by: $L(\lambda) = \prod_{i=1}^N f(x_i; \lambda) = \prod_{i=1}^N \lambda e^{-\lambda x_i}$. The log-likelihood is:

$$\begin{aligned}
 \ln L(\lambda) &= \ln \left[\lambda^N \prod_{i=1}^N e^{-\lambda x_i} \right] \\
 &= \ln(\lambda^N) + \sum_{i=1}^N \ln(e^{-\lambda x_i}) \\
 \ln L(\lambda) &= N \ln(\lambda) - \lambda \sum_{i=1}^N x_i \\
 \frac{\partial \ln L(\lambda)}{\partial \lambda} &= 0 \Leftrightarrow \frac{N}{\lambda} = \sum_{i=1}^N x_i \\
 \hat{\lambda} &= \frac{N}{\lambda} - \sum_{i=1}^N x_i = 0
 \end{aligned}$$

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fe) The minimum variance bound is given by the Cramér-Rao inequality by:

$$V(\lambda) \geq I(\lambda)^{-1}.$$

Where $I(\lambda) = V(S(\lambda)) = \mathbb{E} \left[\left(\frac{\partial \ln L(\lambda)}{\partial \lambda} \right)^2 \right] = -\mathbb{E} \left[\frac{\partial^2 \ln L(\lambda)}{\partial \lambda^2} \right]$ For our case, we have:

$$\begin{aligned}
 \ln L(\lambda) &= \ln \left(\lambda^N \prod_{i=1}^N e^{-\lambda x_i} \right) = N \ln \lambda - \lambda \sum_{i=1}^N x_i \\
 \Rightarrow \frac{\partial \ln L(\lambda)}{\partial \lambda} &= \frac{N}{\lambda} - \sum_{i=1}^N x_i \\
 \Rightarrow \frac{\partial^2 \ln L(\lambda)}{\partial \lambda^2} &= -\frac{N}{\lambda^2} \\
 \Rightarrow I(\lambda) &= -\mathbb{E} \left[\frac{\partial^2 \ln L(\lambda)}{\partial \lambda^2} \right] = +\frac{N}{\lambda^2} \Rightarrow I(\lambda) = \frac{N}{\lambda^2}
 \end{aligned}$$

Thus, at the minimum variance bound,

$$\begin{aligned}
 V(\hat{\lambda}) &= I(\hat{\lambda})^{-1} = \left(\frac{N}{\lambda^2} \right)^{-1} = \frac{\lambda^2}{N} \\
 \Rightarrow V(\hat{\lambda}) &= \frac{\lambda^2}{N}
 \end{aligned}$$

(f) Let's show that the ML estimate of λ is biased and the bias is $b(\hat{\lambda}) = \frac{\lambda^2}{N}$. $\hat{\lambda} = \frac{M}{N} = \frac{N}{N}$.

$$\begin{aligned}
\hat{\lambda} &= \frac{N}{\sum_{i=1}^N x_i} = \frac{N}{s} \text{ with } f_s(s) = \frac{\lambda^N s^{N-1} e^{-\lambda s}}{(N-1)!}, s \geq 0 \\
\Rightarrow \mathbb{E}[\hat{\lambda}] &= \int_0^\infty \hat{\lambda} f_s(s) ds = \int_0^\infty \frac{N}{s} \frac{\lambda^N s^{N-1} e^{-\lambda s}}{(N-1)!} ds \\
&= \frac{N\lambda^N}{(N-1)!} \int_0^\infty \frac{s^{N-1}}{s} e^{-\lambda s} ds = \frac{N\lambda^N}{(N-1)!} \int_0^\infty s^{N-2} e^{-\lambda s} ds \\
&= \frac{N\lambda^{N-1}}{(N-1)!} \int_0^\infty \left(\frac{x}{\lambda}\right)^{N-2} e^{-x} \frac{dx}{\lambda} \quad x = \lambda s \rightarrow s = \frac{1}{\lambda}x \\
&= \frac{N\lambda^{N-1}}{(N-1)! \lambda^{N-1}} \int_0^\infty x^{N-2} e^{-x} dx = \frac{N}{(N-1)!} (N-1)! \lambda \\
\Rightarrow \mathbb{E}[\hat{\lambda}] &= \frac{N}{N-1} \lambda.
\end{aligned}$$